Parag Shende

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Professional Summary

Computer Engineering undergraduate at VIT Pune focused on retrieval-augmented generation, knowledge-graph systems, and applied AI. Designs hybrid retrieval pipelines that outperform single-method baselines and ships full-stack platforms spanning CV/VLM analytics, FastAPI backends, and React frontends. International hackathon winner (1st Prize, Impetus PICT InC 2026). GPA 8.56 / 10.

Education

Vishwakarma Institute of Technology, Pune *August 2023 – June 2027*
B.Tech, Computer Engineering • GPA: 8.56 / 10
Coursework: Software Engineering, Algorithms, Database Management, Object-Oriented Design, Operating Systems, Computer Networks

Technical Skills

Languages: Python, Java, C/C++, SQL, R, Bash
Backend: FastAPI, Flask, Flask-SocketIO, Celery, REST APIs, WebSockets, async I/O
Frontend: React, Next.js 15, HTML5, CSS3, Cytoscape.js
Databases & Stores: PostgreSQL, Redis, MongoDB, Neo4j, FAISS
AI / ML: PyTorch, Hugging Face Transformers, LangChain, RAG (GraphRAG, JsonRAG, Hybrid RAG, Adaptive Hybrid Retrieval), Knowledge Graphs, NVIDIA NIM Embeddings & Reranking, OpenRouter LLMs, BM25, Reciprocal Rank Fusion, Model Context Protocol (MCP), YOLOv8, OpenCV, Whisper, BART, Coqui-TTS
Tools: Git, GitHub Actions, AWS (EC2, RDS), Nginx, pytest, NVIDIA NIM, Ollama, Groq API, GROBID, Swagger/OpenAPI

Experience

SolarKey — Solar Plant Monitoring & O&M Platform *March 2026 – September 2026*
Python, FastAPI, Celery, Next.js 15, PostgreSQL (AWS RDS), Redis, AWS EC2, Nginx

- Built the platform from 0 to 1: created the repository and the initial proof of concept, then took it to a functioning production system served at *api.solar-key.com*, monitoring three live solar plants in Maharashtra at 5-minute telemetry cadence.
- Built the original Node.js backend, then completed its migration (with one colleague) to a single FastAPI + Celery backend over PostgreSQL and Redis, retiring MySQL and InfluxDB; migrated authentication and existing user accounts with passwords intact.
- Built the scheduled data pipeline (telemetry sync, KPI snapshots, daily loss rollups, alarms) after tracing four silent "all clear" dashboard failures to one root cause: tables with readers but no writer.
- Built a 9-tool MCP server exposing the team's root-cause-analysis engine results to LLM clients, preserving explicit refusals (`CANNOT_CHECK`) instead of fabricated values.

Projects

Sanshodhak — Adaptive Hybrid Retrieval RAG for Scholarly Research
Python, FAISS, Neo4j, BM25, NVIDIA NIM, GROBID

- Designed **Adaptive Hybrid Retrieval (AHR)**: a three-channel pipeline fusing BM25, dense FAISS retrieval, and knowledge-graph expansion via Reciprocal Rank Fusion, with slot reservation guaranteeing graph-expanded chunks in the final top-k.
- Reached **NDCG@10 = 0.917** (vs. 0.876 dense-only) and MRR = 0.903 on a 100-question evaluation over a 163-paper corpus, outperforming dense-only, hybrid-only, and graph-only baselines.
- Implemented query expansion, personalized PageRank, and a centrality boost; measured a 6.7-point NDCG@10 regression on the graph-only system and reverted it, tracing the cause to PageRank favouring citation hubs

over the specific paper each question targets.

Sanjaya — Real-Time Video Intelligence Platform **1st Prize, Impetus PICT InC 2026**
Python, FastAPI, React, YOLOv8, Neo4j, FAISS, Hybrid RAG

- Built an end-to-end video analytics system: a YOLOv8-driven CV pipeline for salient frame extraction and detection, with a VLM analyser grounded in CV output to reduce hallucination.
- Structured scene knowledge into a Neo4j knowledge graph and designed Hybrid RAG fusing GraphRAG, JsonRAG, and FAISS dense retrieval for natural-language video queries.
- Delivered the platform via a React frontend on an async FastAPI backend supporting real-time upload, query, and graph visualisation.

Multimodal RAG Pipeline for Medical Transcription & EHR Insights **3rd Place, ASPIRE 2025**
Python, OpenAI Whisper, BART-CNN-Large, Coqui-TTS, RAG

- Built an end-to-end multimodal pipeline chaining Whisper (ASR), fine-tuned BART-CNN-Large (clinical summarization), and Coqui-TTS (voice synthesis) for automated medical transcription and EHR-driven insight generation.
- Implemented Retrieval-Augmented Generation grounded in patient history, medical conditions, and prescription context to deliver context-aware clinical outputs.

Publications

- V. J. Nandeswar, **P. Shende**, et al., "Smart Soil Nutrient Analyzer," *2024 IEEE World Forum on Public Safety Technology (WFPST)*, 2024. IEEE Xplore
- V. Nandeshwar, S. Shelke, M. Shende, V. Shelke, **P. Shende**, R. Shelke, "MediExchange: A Smart Web Portal to Buy and Sell Unused Medicines," *2024 4th International Conference on Sustainable Expert Systems (ICSES)*, IEEE, 2024. IEEE Xplore

Achievements

- **1st Prize** — Impetus, PICT InC 2026 (International-Level Technical Competition); Data Science / AI & ML category, Pune Institute of Computer Technology — for *Sanjaya*.
- **3rd Place** — ASPIRE 2025 National-Level Project Competition (CSI Student Chapter & Society for Data Science); multimodal RAG pipeline for medical transcription and EHR insights.